A compact system of four sub-Neptune-mass planets around HD 32564 in public HARPS velocities

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Abstract

HD 32564 (HIP 23575) is a G6 dwarf of $0.92\,M_\odot$ at 48.5 pc, old and magnetically quiet. HARPS observed it on 193 nights between 2009 and 2016, and its public velocities carry four Keplerian signals, at 2.5566, 11.063, 23.61 and 49.13 d, with semi-amplitudes of 4.2, 3.5, 3.0 and $1.7\,m\,s^{-1}$: four planets with minimum masses of 8.5, 11.4, 12.5 and $9.4\,M_\oplus$ within 0.26 au of the star. Each signal stands in three reductions of the same spectra, SERVAL's velocities, the HARPS pipeline's cross-correlation velocities and line-by-line velocities measured here, the last from all lines and from the lines whose shifts do not follow the Ca II H and K index alike, and each keeps its amplitude and phase from the first half of the nights to the second. None of eight activity indicators has power at any of the four periods. A refit and simulations of the observing window reject the innermost signal's daily alias, whose peak lies at 1.635 d, and every planet's one-year aliases; N-body integrations keep the system stable for 10^6 orbits of the innermost planet on circular orbits, and TESS sees no transits. A fifth signal, at 123.0 d and $1.1\,m\,s^{-1}$, improves the fit by $\Delta BIC = 26$ in SERVAL's velocities and stands in the pipeline's at a false-alarm level of 0.05; it is a candidate, and the five-planet system stays quasi-periodic only if its orbits are close to circular.

Keywords: Techniques: radial velocities, Planets and satellites: detection, Planetary systems, Stars: individual: HD 32564

1. Introduction

Compact systems of planets between the Earth's and Neptune's mass, several of them within a few tenths of an astronomical unit of a Sun-like star, are among the commonest planetary systems in the Galaxy. The transit surveys found them in hundreds, their planets alike in size and evenly spaced [1], their neighbouring periods often a little wide of the low-order resonances [2]; the radial-velocity surveys weigh the nearby ones. HARPS has watched several thousand nearby stars since 2003, and HARPS-RVBank [3] recomputed every public HARPS velocity with SERVAL [4], corrected the instrument's nightly zero points and attached each spectrum's activity indicators, so that each star's whole record can be read at once.

HD 32564 is one of those stars. Its velocities were taken on 193 nights between 2009 and 2016, and no planet has been reported around it. Section 2 describes the data, Section 3 the star, Section 4 the signals and the models that hold them, Section 5 the tests against stellar activity and the instrument, Section 6 the orbits, masses and dynamics, and Section 7 the system's architecture.

2. Data

HARPS-RVBank holds 205 velocities of HD 32564 from 193 nights between 2009 November 29 and 2016 March 18, with a median error of $1.05\,m\,s^{-1}$; seven nights follow HARPS's fibre

upgrade of 2015 June, across which the velocities take an offset of their own. The velocities of each night are averaged, weighted by their inverse variances. The same spectra give two further sets of velocities. DACE holds the HARPS pipeline's cross-correlation velocities, 195 nights after the pipeline's quality control, with the Mount Wilson S index of each spectrum. The third set was measured here from the ESO Science Archive's HARPS spectra, the one of highest signal to noise from each of 199 nights: every spectrum is put on a grid uniform in log wavelength at $0.4\,\mathrm{km\,s^{-1}}$ from 3800 to 6880 Å in the star's frame, the median spectrum is the template, and each spectrum's shift in each of the template's 4 022 lines deeper than 4 per cent is Bouchy's derivative estimate [5] with its photon error. A spectrum's velocity is the weighted mean of its lines, lines more than 5σ from the median set aside and each line carrying a jitter of its own, in the manner of line-by-line velocimetry [6]. Each line's shifts are correlated with the S index measured in the same spectra, and the 303 lines whose correlation passes $p < 0.01$ are left out of a second velocity, made from the 3 719 quiet lines. The activity indicators are the bank's chromatic index, differential line width, Hα and Na D indices and the pipeline's line width and bisector, the S index from DACE and from the line-by-line spectra, and the line-by-line Hα index. The star's TESS light curves [7], sectors 5, 32 and 98 at two-minute or 20-second cadence from SPOC, come through Lightkurve [8].

3. The star

Table 1 gives the star's parameters. Gaia DR3 [9] places it at 48.47 ± 0.04 pc with a renormalised unit weight error of 0.97 and no non-single-star solution, and its astrophysical parameters [10] give a G dwarf of 5 550 K, solar metallicity, $0.92\,\mathrm{M_\odot}$ and $0.96\,\mathrm{R_\odot}$, 8.6 Gyr old. The S index of its 207 HARPS spectra is 0.169 ± 0.004, $\log R'_{\mathrm{HK}} = -4.96$ for $B - V = 0.70$ [11], the activity of an old, slowly rotating star; the activity-rotation relation [12] gives a rotation period near 31 d. In TESS's sectors 5 and 32 its light varies by less than 30 ppm, sector 98 carries the 13-day pattern of the spacecraft's orbit, and no rotation shows in any of them.

Table 1: HD 32564.

Parameter	Value	Source
Gaia DR3	3227308623164447616	[9]
G, $G_{\mathrm{BP}} - G_{\mathrm{RP}}$ (mag)	8.42, 0.89	[9]
Parallax (mas)	20.631 ± 0.017	[9]
RUWE	0.97	[9]
T_{eff} (K)	5 550	[10]
$\log g$	4.40	[10]
[M/H]	−0.06	[10]
Mass ($\mathrm{M_\odot}$)	0.92 ± 0.04	[10]
Radius ($\mathrm{R_\odot}$)	0.957 ± 0.020	[10]
Age (Gyr)	$8.6^{+1.4}_{-2.0}$	[10]
S index	0.169 ± 0.004	HARPS, DACE
$\log R'_{\mathrm{HK}}$	−4.96	this work

4. The signals

4.1. Periodograms

A generalised Lomb-Scargle periodogram [13] of SERVAL's velocities from 1.2 to 3 000 d, with the offset across the fibre upgrade fitted, peaks at 2.557 d at a false-alarm level [14] of 10^{-15}. With that signal's sinusoid fitted, the residuals peak at 11.062 d (4×10^{-18}), then at 23.596 d (10^{-22}), at 49.149 d (7×10^{-13}) and at 123.0 d (10^{-5}); with all five fitted, the highest peak, at 1.393 d, stands at a false-alarm level of 0.2 (Figure 1).

4.2. Models

Keplerian models of three to six planets were fitted by least squares from the periodograms' periods, each with an offset on each side of the fibre upgrade and a jitter added to the errors (Table 2). With every orbit circular, the fourth planet lowers the Bayesian information criterion [15] by 64 and the fifth by a further 26; a sixth, at 1.393 d, lowers it by 4, which does not support it. Free eccentricities improve the log-likelihood of the four-planet model by less than 0.1 and of the five-planet model by 2.0 for ten parameters, so the circular models are preferred.

Table 2: Keplerian models of SERVAL's velocities of HD 32564: log-likelihood with free and with circular orbits, the Bayesian information criterion of each, the free model's jitter and its residual scatter.

Planets	$\ln L$	$\ln L$ (circular)	BIC	BIC (circular)	Jitter ($\mathrm{m\,s^{-1}}$)	Scatter ($\mathrm{m\,s^{-1}}$)
3	−418.1	−420.1	930.9	903.4	1.84	2.11
4	−380.2	−380.2	881.5	839.3	1.41	1.75
5	−357.1	−359.1	861.5	812.9	1.15	1.55
6	−339.8	−349.0	853.3	808.6	0.97	1.43

4.3. The innermost period

The 2.5566-d signal's daily alias, at $|1/P - 1/\mathrm{d}|$, has its peak in SERVAL's velocities at 1.635 d, two cycles a year from the exact alias at 1.650 d, where the window's yearly sidelobes move it; with the other three signals fitted it carries 0.90 of the period's power, 0.681 against 0.759. Refitted with the innermost planet at 1.635 d, the four-planet model loses 31.2 in log-likelihood. The observing window decides as well [16]. Three hundred sets of velocities drawn at the observed times from the four-planet model with the fitted jitter, with the innermost planet at 2.5566 d, give a ratio of the two peaks' powers of 1.17 ± 0.04, and 72 per cent of them reach the observed 1.14; three hundred with the planet at 1.635 d give 0.87 ± 0.03, and none reaches it. The period is 2.5566 d. The one-year aliases of the other three planets, at 10.72 and 11.42, 22.12 and 25.30, and 43.05 and 57.26 d, fit worse by 91, 88, 73, 67, 31 and 29 in log-likelihood.

4.4. The fifth signal

The 123-d signal stands at a false-alarm level of 10^{-5} in SERVAL's velocities with the four planets fitted, and in the pipeline's velocities its peak, at 124.0 d, stands at 0.05. It is a candidate. Fitted with free eccentricities, it takes 0.19 and the 49-d planet 0.20, and the five-planet system then turns chaotic (Section 6).

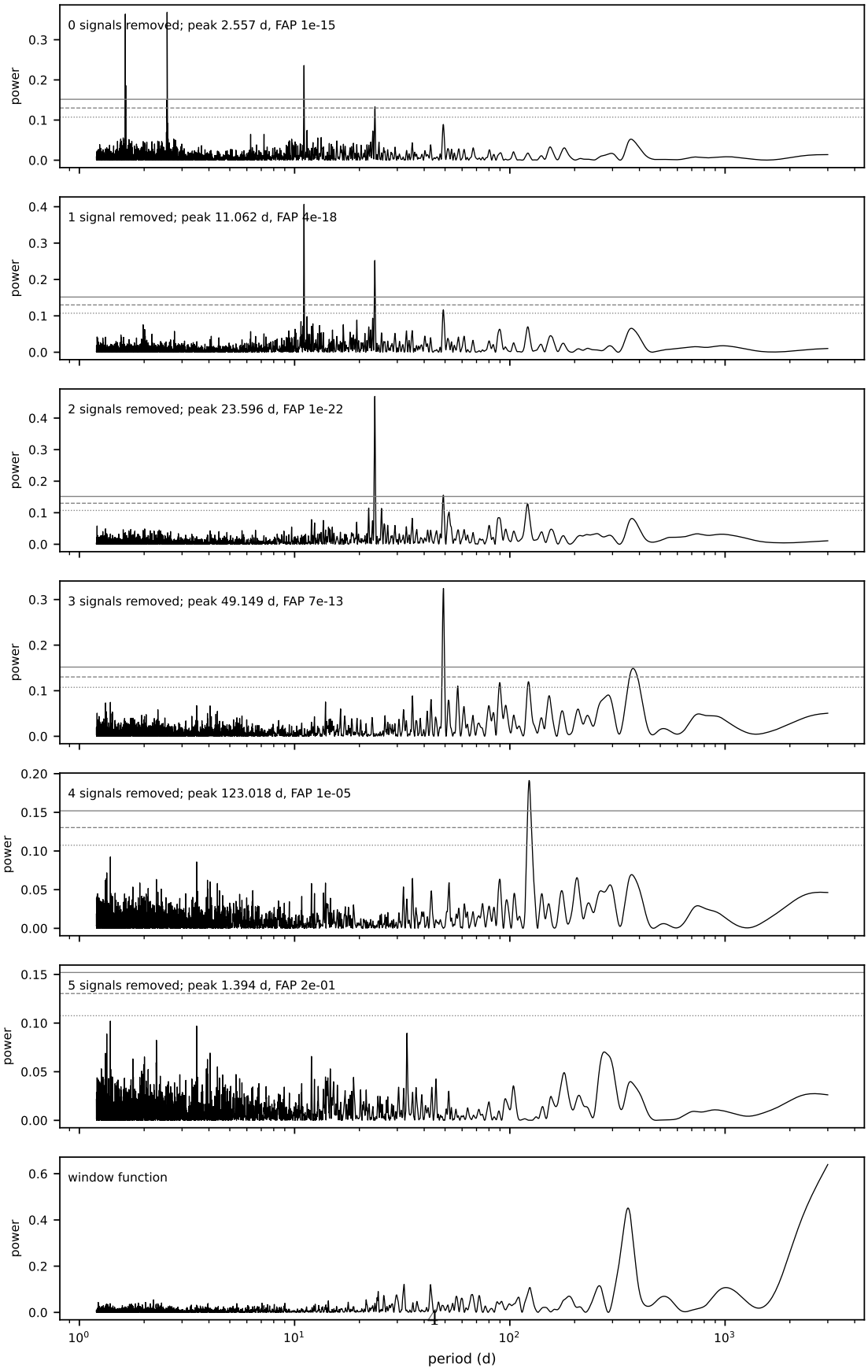


Figure 1: Generalised Lomb-Scargle periodograms of SERVAL's nightly velocities of HD 32564, each with the sinusoids of the signals above it fitted jointly. The horizontal lines are the false-alarm levels of 0.1, 0.01 and 0.001.

5. Activity and the instrument

5.1. Three reductions

Table 3 gives each signal in each of the three reductions, found in the residuals of the other three signals, with its semi-amplitude fitted to the first and to the second half of the nights apart. SERVAL fits a template to every spectrum's whole extracted orders, the pipeline cross-correlates a binary mask, and the line-by-line velocities weigh each line by its own scatter, so a signal one reduction made would not stand in the others; all four stand in all three, with the same phase in the two halves of each and amplitudes that agree within twice their errors. They stand at the same strength in the velocities of the 3 719 lines whose shifts do not follow the S index as in those of all 4 022, as they would not if the activity-sensitive lines carried them.

Table 3: The four signals of HD 32564 in three reductions of its HARPS spectra: each signal's false-alarm level in the residuals of the other three, and its semi-amplitude ($\mathrm{m\,s^{-1}}$) in the first and second half of the nights.

Reduction	2.5566 d	11.063 d	23.61 d	49.2 d
SERVAL (193 nights)	2×10^{-60}	10^{-45}	2×10^{-33}	4×10^{-17}
	4.0, 4.6	3.4, 3.4	2.7, 2.8	1.7, 1.7
Pipeline (195 nights)	10^{-59}	2×10^{-41}	6×10^{-33}	4×10^{-7}
	4.1, 4.5	3.5, 3.2	2.7, 2.9	1.2, 1.3
Line-by-line, all lines (199)	4×10^{-28}	7×10^{-16}	10^{-14}	10^{-5}
	4.1, 4.4	3.1, 3.3	3.1, 3.4	1.7, 2.3
Line-by-line, quiet lines (199)	10^{-26}	9×10^{-15}	10^{-13}	2×10^{-5}
	4.1, 4.2	3.0, 3.3	3.0, 3.4	1.7, 2.4

5.2. Activity indicators

The periodograms of the chromatic index, the differential line width, the Hα and Na D indices, the pipeline's line width and bisector, the S index from DACE's spectra and from the line-by-line spectra have no power at any of the four periods: within 5 per cent of each, no indicator's highest peak stands below a false-alarm level of 0.99. Their own highest peaks lie at a year (Hα and Na D), at four years (the chromatic index) and at the longest periods searched (the line width and bisector), and neither S index has a significant peak at all. The star's rotation, near 31 d, matches none of the periods or their harmonics.

5.3. TESS

Box least squares within 2 per cent of each period finds no transit in the three sectors: at 2.5566 d the best box's 39 events, measured one by one, have a median depth of 35 ppm, where a planet of 9 $\mathrm{M_\oplus}$ and 2.5 to 3 Earth radii would take 600 to 800 ppm from every one. The four planets' transit probabilities are 13, 5, 3 and 2 per cent.

6. Orbits, masses and stability

Table 4 gives the planets' orbits, sampled with emcee [17] and its differential-evolution moves [18] from SERVAL's nightly velocities, each orbit circular, with an offset on each side of the fibre upgrade, a jitter added to the errors and the star's mass of 0.917 $\mathrm{M_\odot}$. Sampled with the fifth signal in the model, the four planets keep their values within their errors, and the fifth

has a period of 122.96 ± 0.69 d, a semi-amplitude of $1.13 \pm 0.17\,\mathrm{m\,s^{-1}}$ and a minimum mass of $8.3 \pm 1.2\,\mathrm{M_\oplus}$ at 0.47 au. Figure 2 shows each planet's orbit with the others' removed.

Table 4: The planets of HD 32564: medians and 68 per cent intervals of the circular orbits sampled from SERVAL's velocities, with the star's mass of $0.917\,\mathrm{M_\odot}$. The jitter is $1.46 \pm 0.10\,\mathrm{m\,s^{-1}}$.

Planet	Period (d)	K ($\mathrm{m\,s^{-1}}$)	$m \sin i$ ($\mathrm{M_\oplus}$)	a (au)
HD 32564 b	2.55660 ± 0.00010	4.22 ± 0.16	8.5 ± 0.3	0.0356
HD 32564 c	11.0634 ± 0.0018	3.46 ± 0.16	11.4 ± 0.5	0.0944
HD 32564 d	23.613 ± 0.011	2.95 ± 0.16	12.5 ± 0.7	0.157
HD 32564 e	49.13 ± 0.07	1.74 ± 0.16	9.4 ± 0.9	0.255

Integrated with REBOUND [19] and its WHFast integrator [20] for 10^6 orbits of the innermost planet, some 7 000 yr, coplanar and edge-on at the minimum masses, the four- and five-planet systems on circular orbits stay bounded with a MEGNO [21] of 2.0, the value of quasi-periodic motion. The four-planet fit with free eccentricities settles on circular orbits. The five-planet fit with them gives the 49-d planet and the fifth signal eccentricities of 0.20 and 0.19, and that system, though bounded over the integration, turns chaotic, its MEGNO reaching 4.5. Adjacent planets on circular orbits are separated by 32, 17 and 17 of their mutual Hill radii, and the fifth signal's orbit by 22 from the fourth.

The four-planet residuals bound what else the velocities hold (Figure 3). A circular orbit is recovered 90 per cent of the time at a semi-amplitude of 1.1 to $1.9\,\mathrm{m\,s^{-1}}$ out to 200 d, a minimum mass of 2.4 to $3.0\,\mathrm{M_\oplus}$ at 1.5 to 2.7 d, 3.4 to $4.1\,\mathrm{M_\oplus}$ at 3 to 9 d, 4.6 to $7\,\mathrm{M_\oplus}$ at 10 to 34 d, 7.3 to $8.9\,\mathrm{M_\oplus}$ at 41 to 74 d and 10 to $14\,\mathrm{M_\oplus}$ at 90 to 200 d. From 240 to 520 d the limit lies at 22 to $33\,\mathrm{M_\oplus}$, near 430 d at $99\,\mathrm{M_\oplus}$, and beyond 600 d no semi-amplitude up to $15\,\mathrm{m\,s^{-1}}$ is recovered 90 per cent of the time.

7. Discussion

HD 32564's four planets are alike: their minimum masses lie between 8.5 and $12.5\,\mathrm{M_\oplus}$, and they sit at 0.036, 0.094, 0.157 and 0.255 au from a star of 0.78 solar luminosities, where they receive 620, 88, 32 and 12 times the Earth's insolation. The outer three are spaced nearly evenly in the logarithm of their periods, at ratios of 2.13 and 2.08, each a few per cent wide of the 2:1 resonance, as neighbouring planets in the transiting multi-planet systems most often are [2]; the alikeness of masses and spacing is the pattern those systems show in their sizes [1]. The ratio between the innermost two, 4.33, is the system's widest: a planet midway, near 5.3 d, would continue the spacing of the outer three, and the velocities hold none above $3.8\,\mathrm{M_\oplus}$ there. The fifth signal, at 123.0 d, would extend the sequence at a ratio of 2.50.

With minimum masses of 8.5 to $12.5\,\mathrm{M_\oplus}$, the planets are heavier than the Earth-like rocky planets and lie where sub-Neptunes with gaseous envelopes are common. None transits for TESS; the innermost planet's transit probability is 13 per cent, and in a coplanar system the outer planets, at 5, 3 and 2 per cent, could transit only where it does.

8. Conclusion

HD 32564, a quiet G6 dwarf at 48.5 pc, hosts four planets of 8.5 to $12.5\,\mathrm{M_\oplus}$ on orbits of 2.5566, 11.063, 23.61 and 49.1 d, read from 193 nights of public HARPS velocities. Each planet stands in

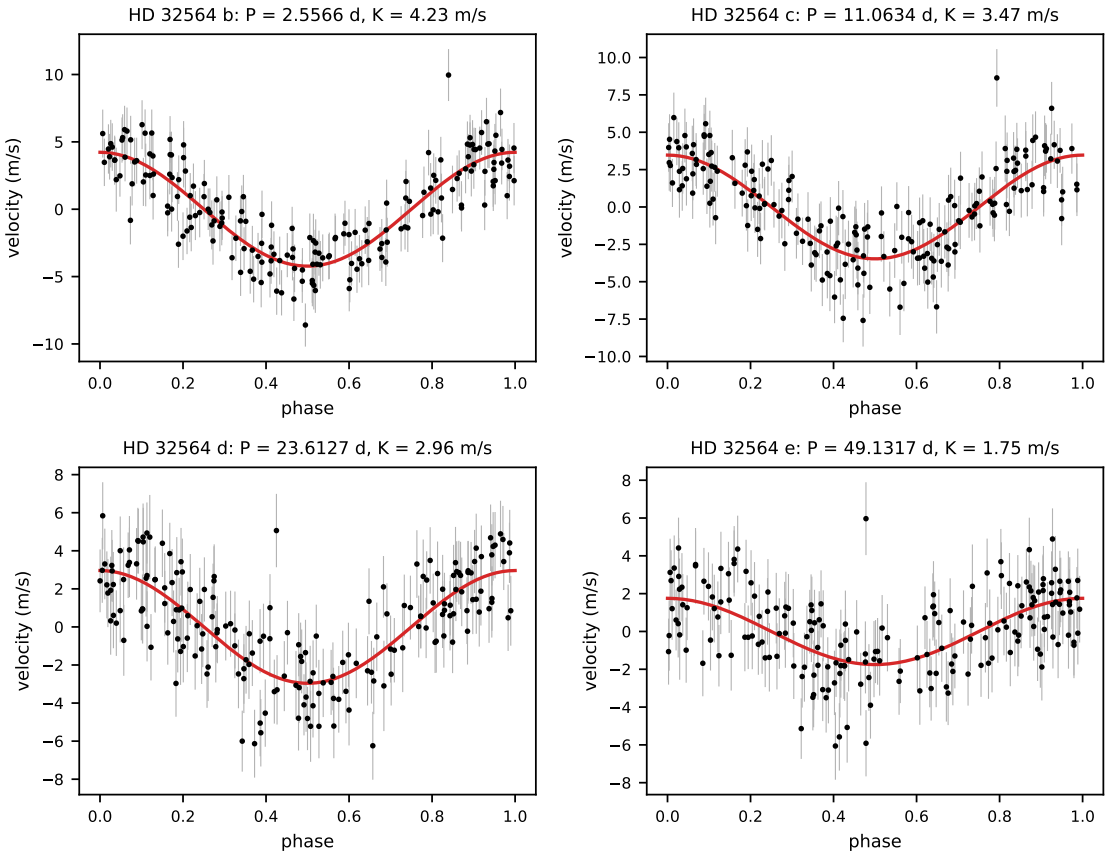


Figure 2: SERVAL's nightly velocities of HD 32564 phased at each planet's period, the other planets' orbits removed, with the circular orbit.

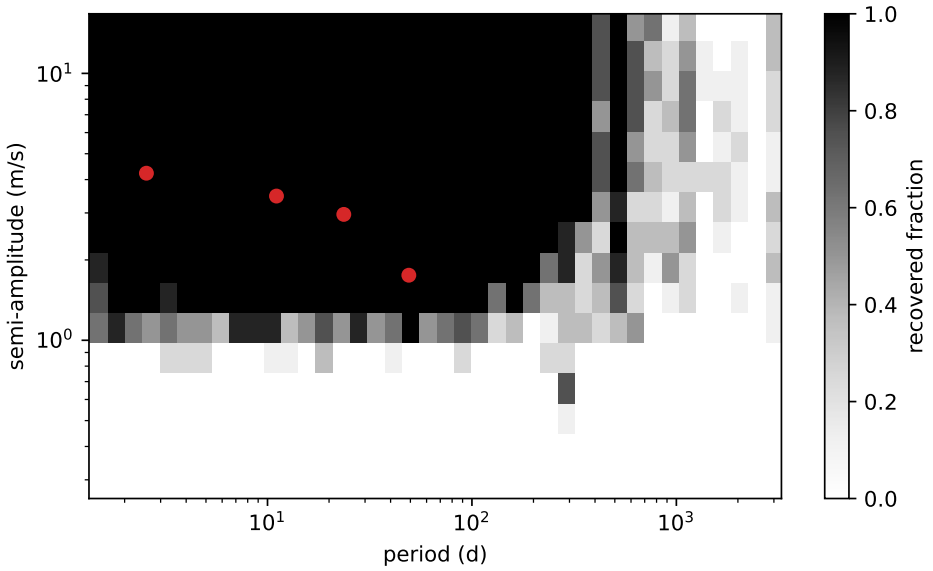


Figure 3: The fraction of circular orbits injected into the four-planet residuals that are recovered, the highest peak of the residual periodogram within 2 per cent of the injected period at a false-alarm level below 10^{-3}, eight phases at each period and semi-amplitude. The four planets are marked.

three reductions of the same spectra and in the activity-quiet lines alone, keeps its amplitude and phase across the data, and has no counterpart in eight activity indicators; the innermost period is the true one against its daily alias, and the system is stable on circular orbits. A fifth signal at 123.0 d is a candidate. Velocities from another instrument would test it, and would measure the four planets' eccentricities, which the dynamics already hold close to zero.

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